Microeconomic Theory: TA Session

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Reminder

No TA sessions and office hours next week due to Spring Break

Perfect competition

In a perfectly competitive market, buyers and sellers are price takers

Characteristics of a perfectly competitive market:

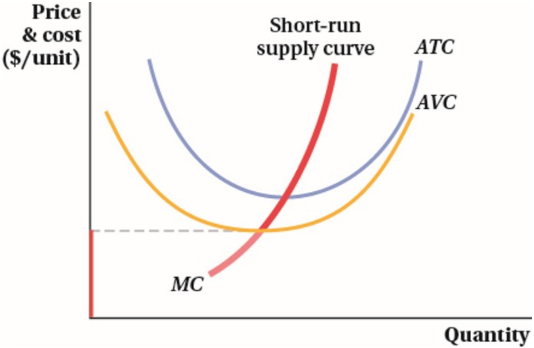
- The market has many buyers and sellers
- The goods offered by sellers are identical
- Firms can freely enter or exit the market

Short-run supply curve of a firm

Recall that firm will produce where P=MC, so the firm's supply curve overlaps with MC curve

But the firm will shut down if $P < AVC$

The short-run supply curve is the part of the MC curve above AVC



The long-run decision to exit or enter

From previous weeks, we know that a firm chooses to exit if:

$$\implies P < ATC$$

On the contrary, a firm chooses to enter the market if:

$$P > ATC$$

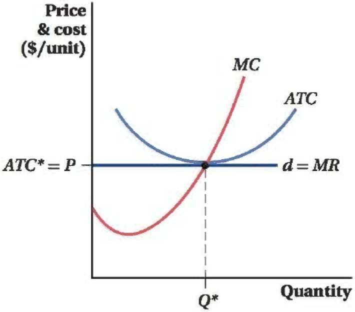
If profits are positive, more firms will enter the market; if profits are negative, existing firms will exit the market

In equilibrium, all firms make zero profit in the long run: $P = ATC$

Zero profit in the long run

In equilibrium, all firms make zero profit in the long run: $P = ATC$

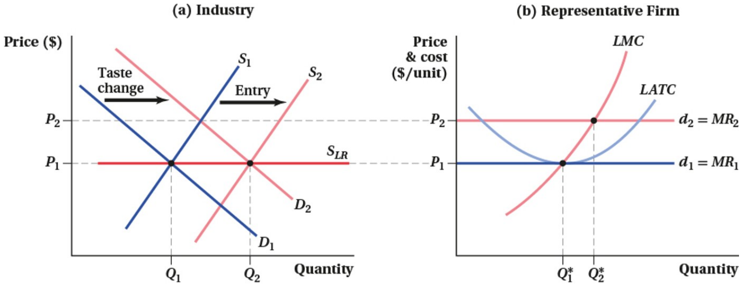
- Recall that $MC = ATC$ where ATC is minimized
- Profit maximization requires $P = MC$
- $\Rightarrow$ In the long-run equilibrium, $P = MC = ATC$, firm's production happens where ATC is minimized



The effect of a change in demand in the long run

In the following, we consider the simple scenario where all firms are identical and their cost curves don't change

Suppose a change in taste increases demand from D_1 to D_2
$\Rightarrow$ Short-run price increases from P_1 to P_2
$\Rightarrow$ Existing firms make a profit, new firms enter the market
$\Rightarrow$ The entry of new firms increases supply (from S_1 to S_2) until all firms make zero profit
$\Rightarrow$ The new equilibrium is characterized by D_2 and S_2, quantity increases from Q_1 to Q_2, price remain unchanged at P_1, which is also the representative firm's lowest ATC



Industry under perfect competition - exercise

A competitive firm has cost function $TC = q^2 + 10q + 100$ and faces a market demand $Q = 1000 - 20P$.

1. Derive the firm's MC and ATC.

   Answer: $MC = 2q + 10, \;\; ATC = q + 10 + \frac{100}{q}$.

2. If there are currently 60 identical firms, find the short-run equilibrium price and quantity.

   Answer: With 60 identical firms,
   $60q = 1000 - 20P \;\Rightarrow\; P = 50 - 3q$. Using $P = MC$, we have
   $50 - 3q = 2q + 10 \;\Rightarrow\; q = 8, \;\; P = 50 - 3 \times 8 = 26$

3. Determine whether firms are making profit or loss, and how much, at this equilibrium.

   Answer: At $q = 8, \;\; TC = q^2 + 10q + 100 = 244$. Total revenue $TR = Pq = 26 \times 8 = 208$. $TR < TC$, each firm makes a loss of 36.

Industry under perfect competition - exercise

A competitive firm has cost function $TC = q^2 + 10q + 100$ and faces a market demand $Q = 1000 - 20P$.

4. What will happen in the long run? Compute the long-run equilibrium price, quantity, and the number of firms.

   Answer:

   Suppose there are n firms. Then, $nq = 1000 - 20P \;\Rightarrow\; P = 50 - \frac{nq}{20}$

   Profit maximization $\Rightarrow\; P = MC \;\Rightarrow\; 50 - \frac{nq}{20} = 2q + 10.$

   Zero profit $\Rightarrow\; P = ATC \;\Rightarrow\; 50 - \frac{nq}{20} = q + 10 + \frac{100}{q}.$

   Then we can solve $MC = ATC$ to directly get q:
   $2q + 10 = q + 10 + \frac{100}{q} \;\Rightarrow\; q = 10.$

   Using $P = MC$ again, we get $P = 2q + 10 = 30.$

   Finally, to solve n, we use $P = 50 - \frac{nq}{20} = 30 \;\Rightarrow\; n = 40.$

Industry under perfect competition - exercise

A competitive firm has cost function $TC = q^2 + 10q + 100$ and faces a market demand $Q = 1000 - 20P$.

5. Suppose the industry is now at the long-run equilibrium with 40 firms. The government now offers firms a subsidy of \$10 per unit. Calculate the short-run quantity, price and profit of each firm.

   Answer: The subsidy reduces firms' cost by \$10 per unit of q. The new $TC = (q^2 + 10q + 100) - 10q = q^2 + 100$, $MC = 2q$.

   $P = MC: \ 50 - \frac{nq}{20} = 50 - 2q = 2q \ \Rightarrow \ q = 12.5, \ P = 25.$

   $\text{Profit} = PQ - TC = 12.5 \times 25 - (12.5^2 + 100) = 56.25.$

6. Calculate the long-run quantity, price, and number of firms.

   Answer: In the long run, $P = MC = ATC$.

   $MC = ATC: \ 2q = q + \frac{100}{q} \ \Rightarrow \ q = 10.$

   $P = MC: \ P = 2q = 20.$

   Suppose the number of firms is n.

   $P = 50 - \frac{nq}{20} = 20 \ \Rightarrow \ n = 60.$